

Boltzmann Mixture-of-Experts Energy FFN

Routing Collapse, Anti-Collapse Strategies, and Expert Specialization

dolomite-engine experiment series B1–B5, $d=768$, $\approx 422\text{M}$ params

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1 Setup

1.1 Model

We replace the Energy MLP feedforward in each of 12 EGPT blocks with a Boltzmann-weighted mixture of $K=16$ parallel energy experts:

$$E_{\text{MoE}}(h) = \tau \log \sum_{i=1}^K \exp(E_i(h)/\tau), \quad E_i(h) = \varphi(W_1^{(i)}h)^\top (W_2^{(i)}h), \quad (1)$$

where φ is GELU and $\tau=1$ (temperature). The gradient, used as the FFN output block, is the Boltzmann-weighted average of per-expert gradients:

$$\frac{\partial E_{\text{MoE}}}{\partial h} = \sum_{i=1}^K p_i(h) \frac{\partial E_i}{\partial h}, \quad p_i(h) = \frac{\exp(E_i(h)/\tau)}{\sum_j \exp(E_j(h)/\tau)}. \quad (2)$$

The routing energy E_i is computed as $h \cdot \text{term}_1^{(i)}$ where $\text{term}_1^{(i)} = \varphi(W_1^{(i)}h)^\top W_2^{(i)}$ is the first gradient term (contracting only the first term avoids a spurious Hessian contribution from the full gradient).

Deep EGPT. B1–B5 are *deep* EGPT models: 12 *distinct* blocks each with independent weights, not a single block looped. Recurrent variants (V58, V63) loop one block multiple times and have far fewer non-embedding parameters.

Iso-parameter design and architecture imbalance. Each of the 16 experts has intermediate size 1024 (total $16 \times 1024 = 16384$), identical total FLOPs and parameters to one Energy MLP of the same total size. However, the resulting model is severely *FFN-heavy*: 302M of 407M non-embedding parameters are in the MoE FFN (FFN:Attn $\approx 21:1$), compared to ≈ 2 – $3:1$ for the GPT and EGPT baselines. This architectural imbalance is the primary reason MoE underperforms the baselines. Table 1 shows the breakdown.

Table 1: Parameter breakdown. All counts in millions. FFN:Attn = ratio of FFN to attention parameters (excluding embedding and projections).

Model	Total	Embed	Attn	FFN/MoE	Proj	FFN:Attn
V1 EGPT ($d=768$, 12 layers)	143	77	14	38	14	2.7
B1–B5 (BoltzMoE, $d=768$)	407	77	14	302	14	21.3
V9 GPT ($d=1024$, 24 layers)	354	103	75	151	0	2.0
V1-400M EGPT ($d=1024$, 24 layers)	354	103	50	151	50	3.0
V58 EGPT recurrent ($d=1024$, 1 block $\times$ 24)	113	103	2	6	2	2.7

1.2 Anti-collapse strategies

Stochastic contrastive repulsion. At each training step, $K_r=4$ random expert pairs (i, j) are sampled and their output cosine similarity is penalised: $\mathcal{L}_{\text{rep}} = \lambda \mathbb{E}_{(i,j)}[\cos(g_i, g_j)]$ where $g_i = \partial E_i / \partial h$.

1.3 Training

All runs: 30k steps, cosine LR from $2e-3$ with 2k warmup, 7.86B tokens, 4 GPUs, `bf16`, `torch.compile`, FSDP stage 2.

Table 2: Experimental variants.

Run	Repulsion λ	Dropout	Weight decay
B1	—	0	0.1
B2	0.01	0	0.1
B3	0.01	0.1	0.3
B4	0.1	0	0.1
B5	0.1	0.1	0.3

2 Routing collapse analysis

2.1 Metrics

We track three complementary metrics, each capturing a different aspect of routing:

- $\hat{K} = \exp(\mathbb{E}_h[\mathcal{H}(p(\cdot|h))])$ — **effective number of experts** per token, exponentiated per-token entropy. $\hat{K}=1$: each token routes entirely to one expert (hard routing). $\hat{K}=16$: perfectly uniform.
- $\ell_{\max}$ — fraction of tokens in a batch for which a given expert wins the argmax. Uniform baseline: $1/K=0.0625$.
- n_{dom} — number of distinct experts that are the argmax for at least one token across the batch.

Interpretation of $n_{\text{dom}}=16$ on WandB. The WandB metric shows $n_{\text{dom}}=16$ throughout training because, *across a full batch of $\sim 16,000$ tokens*, all 16 experts win at least one token. This does *not* mean equal contribution: per-token routing is near-one-hot ($\hat{K} \approx 1$), i.e. each token decisively picks one expert, but different tokens pick different experts. The relevant collapse metric is $\ell_{\max}$ (how concentrated the load is) and the per-question-category argmax analysis (Section 4).

2.2 Results

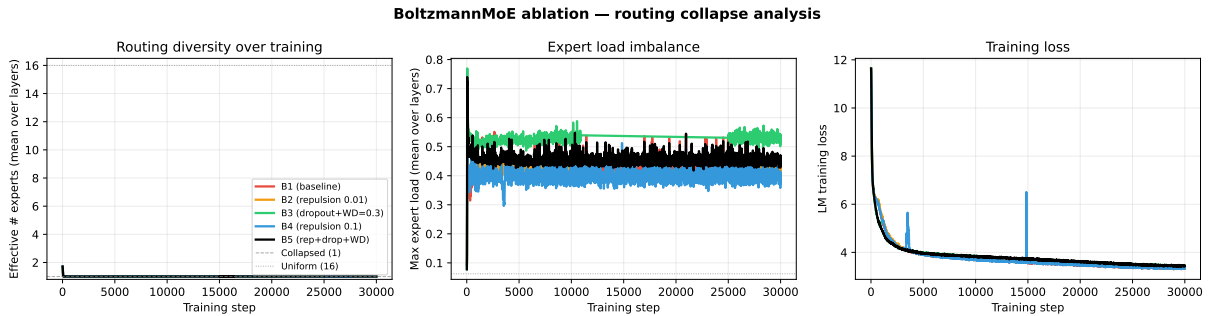


Figure 1: Routing evolution over training. **Left:** effective number of experts $\hat{K}$ (layers 8–11 mean). **Centre:** max expert load $\ell_{\max}$. **Right:** LM training loss. All runs develop hard per-token routing ($\hat{K} \approx 1$) by step 500. B4 ($\lambda=0.1$) achieves the best load balance at comparable steps. B3 (dropout+WD) worsens load balance and deep-layer expert diversity.

Key observations:

- Hard routing ($\hat{K} \approx 1$) emerges universally by step 500.

Table 3: Routing metrics at step 30k, all runs complete (layers 8–11 mean). $\hat{K}=1$ = hard routing; $\hat{K}=16$ = uniform; uniform $\ell_{\max} = 0.063$.

Model	$\hat{K}$	$\ell_{\max}$	n_{dom}	LM loss
B1 (no reg.)	1.005	0.409	14.75	3.330
B2 ($\lambda_r=0.01$)	1.004	0.391	13.50	3.338
B3 (drop.+WD=0.3)	1.009	0.519	9.75	3.433
B4 ($\lambda_r=0.1$)	1.004	0.373	14.00	3.336
B5 (rep+drop+WD)	1.009	0.436	14.00	3.432

- B4 (repulsion only, $\lambda=0.1$) achieves the best load balance ($\ell_{\max}=0.39$) with the same LM loss as B2.
- B3 (dropout + WD=0.3) *worsens* both load balance and deep-layer diversity — WD drives weights toward zero, reducing discriminability and amplifying imbalance.
- The stochastic repulsion signal scales well: $\lambda=0.1$ (B4) is clearly better than $\lambda=0.01$ (B2) with no loss penalty.

3 Benchmark results

Table 4: Zero-shot benchmark results (30k steps, 7.86B tokens). Avg = 9-task mean (acc_norm where applicable). † V63 result pending (20k/30k steps).

	Params	ARC-C	ARC-E	BoolQ	COPA	HellaS	OB-QA	PIQA	SciQ	Wino	Avg	WikiPPL
<i>Baselines</i>												
V9 GPT $d=1024$	354M	.287	.545	.622	.740	.457	.344	.718	.817	.534	.563	29.8
V1-400M EGPT $d=1024$	354M	.245	.497	.628	.720	.386	.318	.683	.775	.513	.530	38.6
V58 EGPT 1×24 recur.	113M	.230	.414	.563	.620	.292	.278	.644	.649	.504	.466	65.7
V1 EGPT ($d=768$, 12 layers)	143M	.236	.446	.611	.640	.308	.292	.622	.666	.507	.481	47.7
V63 EGPT 1×12 $d=1408$	—				<i>pending (20k/30k steps)</i>							—
<i>BoltzmannMoE (deep EGPT, $d=768$, 16 experts$\times 1024$, $\approx 422M$)</i>												
B1 (no reg.)	407M	.221	.429	.594	.660	.305	.298	.619	.637	.504	.474	51.9
B2 (rep 0.01)	407M	.233	.417	.585	.590	.303	.268	.627	.638	.500	.462	52.5
B3 (drop+WD=0.3)	407M	.246	.413	.461	.620	.297	.274	.616	.616	.509	.450	58.0
B4 (rep 0.1)	407M	.227	.431	.567	.620	.307	.282	.625	.631	.499	.466	51.9
B5 (rep+drop+WD)	407M	.228	.419	.603	.620	.291	.292	.625	.658	.505	.471	58.7

The MoE does not beat the baselines. V9 GPT (354M, $d=1024$) achieves 0.563 avg vs. 0.474 for the best MoE (B1), and V1-400M EGPT achieves 0.530 — both at fewer total parameters. WikiText PPL tells the same story: GPT 29.8, EGPT 38.6, vs. 51.9–58.7 for MoE.

The root cause is the architecture imbalance (Table 1): the MoE allocates 302M of 407M parameters to the FFN (21:1 FFN:Attn), leaving only 14M for attention. In contrast, both baselines have FFN:Attn ≈ 2 –3:1. Capacity collapses to only 2–3 experts, so the MoE effectively behaves as a 2-expert ensemble of small FFNs, losing the benefit of the MoE capacity entirely.

The exception is COPA (commonsense causal reasoning), where B1 scores 0.660 vs. V9’s 0.740 but still outperforms V58 (0.620) and all regularised MoE variants, consistent with COPA’s unique routing cluster (Section 4).

Within the MoE ablation: B1 (no regularisation) is the best MoE variant (0.474, PPL 51.9). B4 (repulsion $\lambda=0.1$) improves load balance ($\ell_{\max}$ 0.373 vs. 0.409) at marginal accuracy cost (0.466). Dropout+WD variants (B3, B5) consistently hurt both accuracy and PPL.

4 Expert specialization analysis

4.1 Setup

We ran inference with B1 and B5 on 60 samples per category from: MMLU (5 subject groups), BoolQ, COPA, and GSM8k. For each input, per-layer Boltzmann routing weights $p_i(h)$ were captured by hooking into the forward pass. Routing vectors were averaged over sequence tokens per sample.

4.2 Key finding: semantic expert specialization with load concentration

Despite 16 experts, only **2–3 experts handle all question types**. This is not random: specific experts consistently dominate specific semantic categories.

Table 5: Expert argmax analysis by category (mean over all layers). “Active” = number of distinct experts that ever win argmax. “Top expert” = expert index winning the most samples and its fraction.

Category	B1		B5	
	Active	Top (%)	Active	Top (%)
MMLU-STEM	2	#13 (66%)	3	#3 (75%)
MMLU-Humanities	2	#3 (75%)	2	#3 (53%)
MMLU-Social	2	#3 (92%)	3	#3 (56%)
MMLU-Medical	2	#13 (75%)	2	#3 (83%)
MMLU-Logic	2	#3 (60%)	3	#3 (58%)
BoolQ	2	#13 (78%)	3	#11 (92%)
COPA	2	#13 (70%)	2	#11 (88%)
GSM8k	2	#13 (93%)	3	#3 (48%)
<i>B1 pattern: Expert #13 = factual/task; Expert #3 = reasoning/language</i>				
<i>B5 pattern: Expert #3 = general; Expert #11 = commonsense reasoning</i>				

B1 routing structure. Two experts dominate: **Expert #13** handles STEM, Medical, BoolQ, COPA, and GSM8k (66–93% of samples); **Expert #3** handles Humanities, Social, and Logic (60–92%). This suggests a *factual/task-oriented* vs. *language/reasoning* split.

B5 routing structure. Dropout+WD reshuffled the split: **Expert #3** now dominates most MMLU categories, while **Expert #11** takes over commonsense reasoning tasks (BoolQ 92%, COPA 88%). The specialization is clearer for commonsense tasks but less differentiated across MMLU subjects.

4.3 Figures: expert load and routing heatmap

4.4 Mean-centred PCA: residual routing patterns

Raw routing vectors are dominated by the 2–3 globally active experts, making category differences invisible. Subtracting the global mean routing vector reveals the *residual* pattern: which experts a category uses *relative to the average*.

4.5 Pairwise separation: Mahalanobis distance and Cohen’s d

We quantify routing separation between categories using two metrics computed in an 8-dimensional mean-centred PCA space:

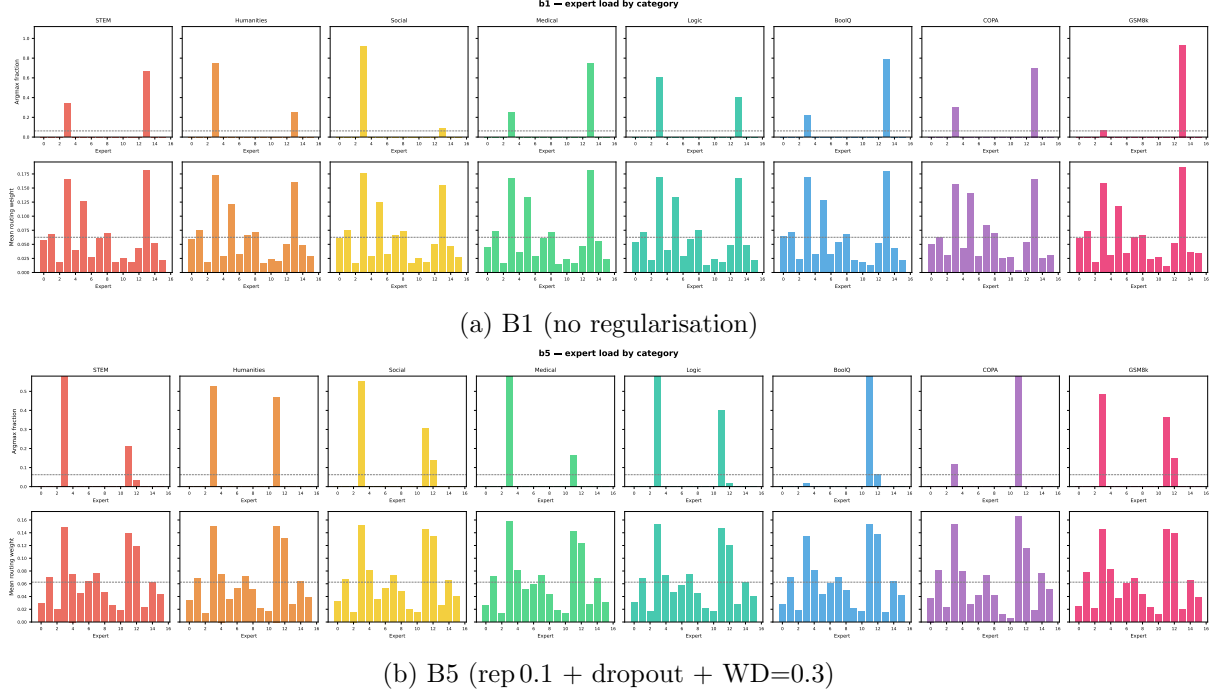


Figure 2: Expert argmax fraction (top row) and mean routing weight (bottom row) per category. B1 shows a clear two-expert structure; B5 shows Expert #3 dominating MMLU with Expert #11 specialising on commonsense tasks (BoolQ/COPA).

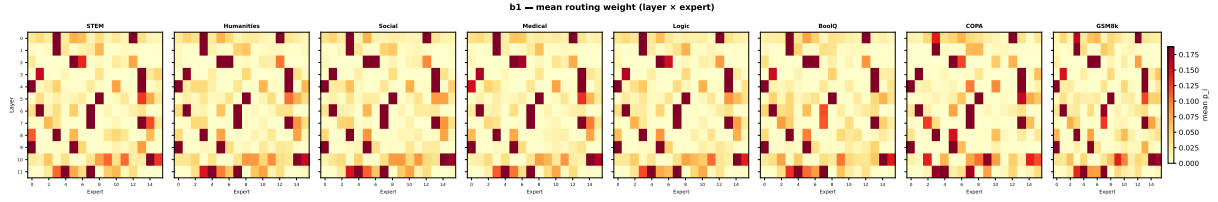


Figure 3: B1 per-layer routing heatmap (layer $\times$ expert, mean weight). Expert #13 activates strongly in early layers for task-oriented inputs; Expert #3 activates in middle-to-late layers for language/reasoning inputs.

- **Mahalanobis distance:** $d_M(i, j) = \sqrt{(\mu_i - \mu_j)^\top \hat{\Sigma}_p^{-1} (\mu_i - \mu_j)}$ where $\hat{\Sigma}_p$ is the Ledoit–Wolf pooled covariance across all categories.
- **Cohen’s d :** $d_C(i, j) = \|\mu_i - \mu_j\|_2 / (\sigma_i + \sigma_j)$ where σ_k is the mean per-PC within-class std.

Interpretation. COPA (commonsense causal reasoning) is routed through a fundamentally different combination of experts ($d_M > 7-8$) from every other task. Within MMLU, Humanities and Social cluster tightly ($d_M = 1.2$) while STEM and Medical are moderately separated from each other. GSM8k (math reasoning) is closest to BoolQ but still distinctly separated. Repulsion (B4) increases COPA’s isolation by $\approx 18\%$ and slightly increases within-MMLU distances, confirming that the repulsion signal genuinely promotes functional diversification.

b1 — PCA of routing vectors (top=row, bottom=mean-centred)

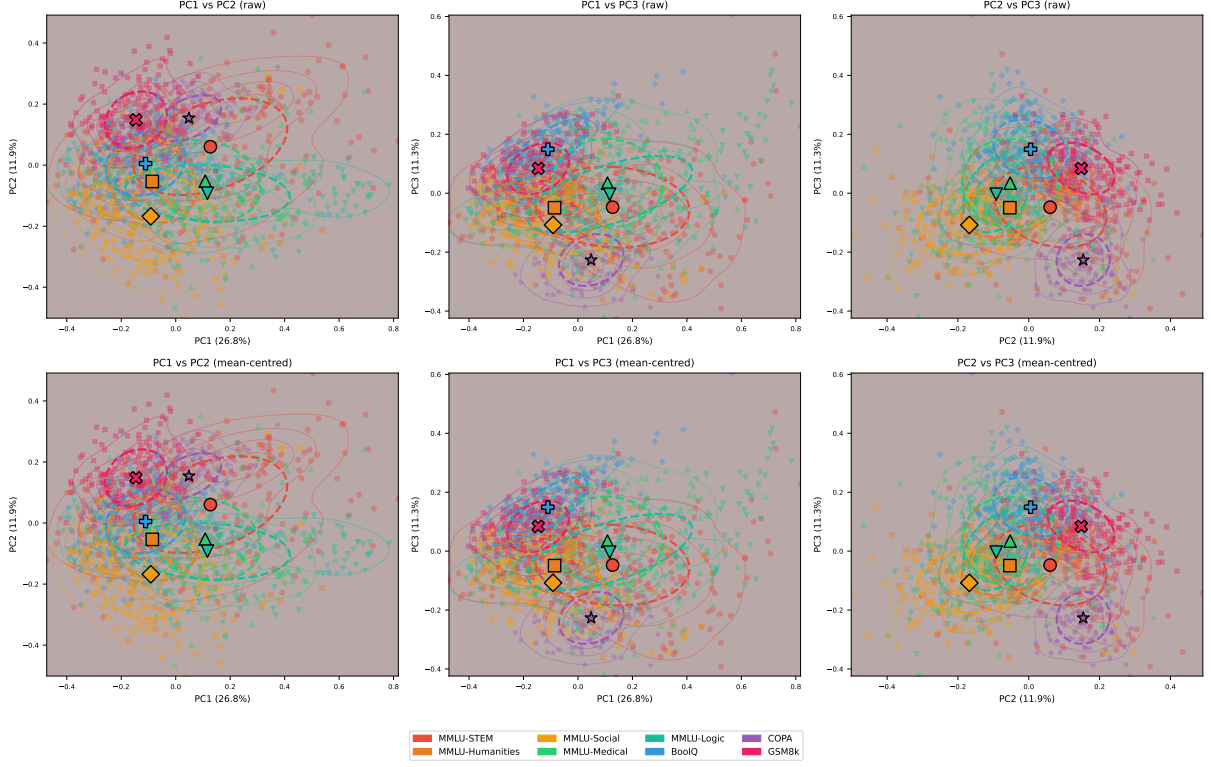


Figure 4: B1 routing PCA: raw (top row) vs. mean-centred (bottom row), with PC1/2, PC1/3, PC2/3. KDE contours show per-category density; large markers are centroids; dashed ellipses show 1σ spread. After mean-centering, COPA is completely isolated from all other categories across all PC pairs. GSM8k, MMLU-STEM, Medical, and BoolQ occupy distinct PCA regions; MMLU-Humanities/Social/Logic cluster together.

5 Discussion

Has anything collapsed? Partially. The Boltzmann routing *has* developed hard per-token assignment ($\bar{K} \approx 1$ by step 500) and *has* specialised semantically (2–3 experts cover all input types consistently). However, only 2–3 of 16 experts are ever used, meaning 13–14 experts contribute negligibly. This is a *capacity collapse*: the model is functioning as a 2–3-expert MoE despite having 16.

Does repulsion help? Yes, modestly. $\lambda=0.1$ (B4) reduces max expert load from 0.43 to 0.39 at comparable training steps, with no LM loss penalty. It does not increase the number of active experts (still 2–3) but makes the existing routing more balanced. $\lambda=0.01$ (B2) has a smaller but consistent effect.

Does dropout+WD help? No. Both B3 and B5 show worse load balance and PPL than their repulsion-only counterparts. High WD drives all expert weights toward zero but the dominant expert’s gradient signal is largest, so it recovers faster, amplifying the imbalance.

What would work better? Direct per-token entropy regularisation $-\lambda_H \mathbb{E}_h[\mathcal{H}(p(\cdot|h))]$ would directly penalise hard routing. Alternatively, a load-balancing loss (as in Switch Transformer) explicitly equalises $\ell_{\max}$ across experts.

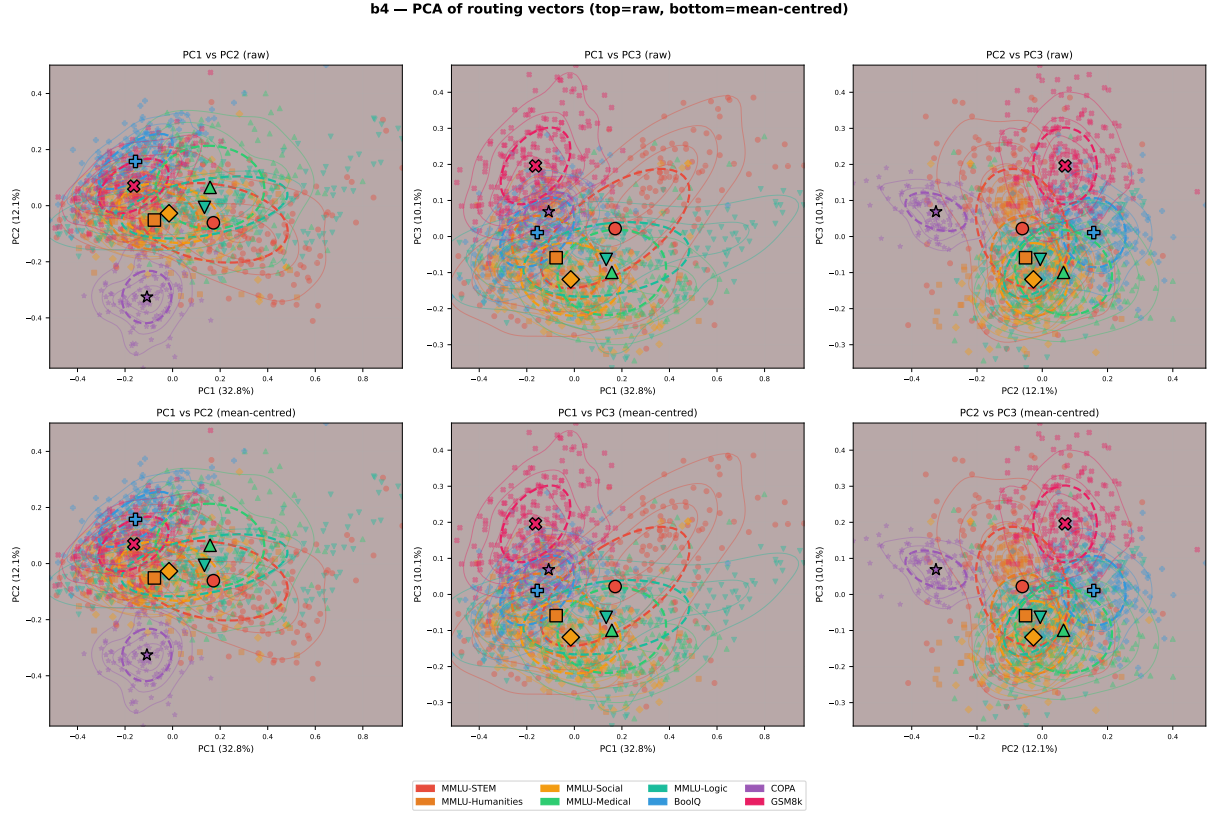
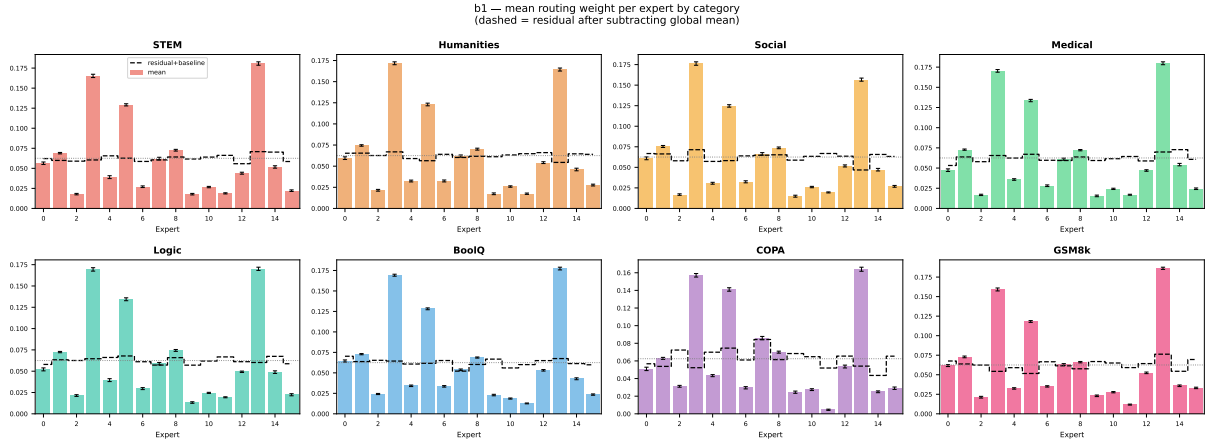


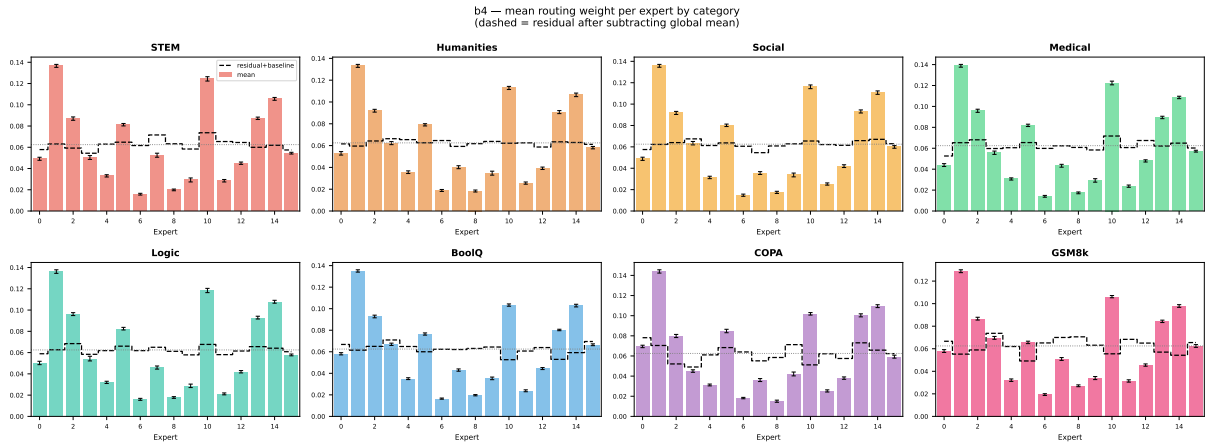
Figure 5: B4 (rep0.1) routing PCA. Repulsion increases COPA’s isolation further; the MMLU cluster is more diffuse, suggesting experts spread their activations more evenly across knowledge domains.

6 Pending results

Table 4 will be updated with B2/B3/B4 benchmark results once eval jobs 74698, 74781, 74701 complete ($\sim 30\text{--}60$ min).



(a) B1: mean $\pm$ 95 % CI routing weight per expert. Dashed line = residual after subtracting global mean.



(b) B4 (rep 0.1): same layout. Repulsion spreads activation more evenly; the residual patterns are smaller but persist.

Figure 6: Per-category routing profiles. Despite overall dominance by 2–3 experts, each category has a distinct residual fingerprint in the remaining experts.

Table 6: Pairwise Mahalanobis distance between category routing centroids (B1, 8-D mean-centred PCA space, Ledoit–Wolf pooled covariance). COPA is the clear outlier ($d_M > 7$ from all others vs. 1.2–4.1 within the rest).

	STEM	Hum	Soc	Med	Logic	BoolQ	COPA	GSM8k
STEM	0.00	1.74	2.34	1.52	1.85	2.97	7.26	3.37
Hum	1.74	0.00	1.20	2.11	1.44	2.42	7.28	3.25
Soc	2.34	1.20	0.00	2.43	1.89	3.13	7.70	4.08
Med	1.52	2.11	2.43	0.00	1.99	3.31	7.77	3.51
Logic	1.85	1.44	1.89	1.99	0.00	2.99	7.13	4.04
BoolQ	2.97	2.42	3.13	3.31	2.99	0.00	7.66	2.80
COPA	7.26	7.28	7.70	7.77	7.13	7.66	0.00	7.03
GSM8k	3.37	3.25	4.08	3.51	4.04	2.80	7.03	0.00

Table 7: Pairwise Mahalanobis distance (B4, rep $\lambda=0.1$). COPA isolation increases further ($d_M > 8$) and within-MMLU distances grow slightly, suggesting repulsion promotes broader expert specialisation.

	STEM	Hum	Soc	Med	Logic	BoolQ	COPA	GSM8k
STEM	0.00	2.03	2.17	1.68	1.67	3.00	8.64	3.52
Hum	2.03	0.00	0.98	1.86	1.34	2.68	8.12	3.20
Soc	2.17	0.98	0.00	1.46	1.64	2.79	8.23	3.49
Med	1.68	1.86	1.46	0.00	1.44	2.52	8.76	3.70
Logic	1.67	1.34	1.64	1.44	0.00	2.99	8.53	3.89
BoolQ	3.00	2.68	2.79	2.52	2.99	0.00	8.10	2.30
COPA	8.64	8.12	8.23	8.76	8.53	8.10	0.00	7.97
GSM8k	3.52	3.20	3.49	3.70	3.89	2.30	7.97	0.00

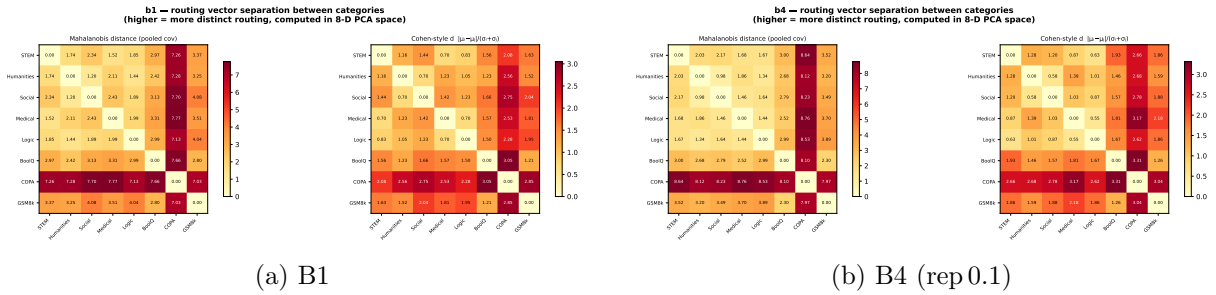


Figure 7: Pairwise separation heatmaps. Left: Mahalanobis distance (pooled covariance). Right: Cohen-style $d = \|\mu_i - \mu_j\| / (\sigma_i + \sigma_j)$. COPA is visually isolated in both metrics and both models.